

IMAT MOCK EXAM 4 — "Calculation Spine"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 43/90.0 · Mock exam

Section A - Logical Reasoning & Reading (Q1-9)

1. "Osmosis is the net movement of water across a selectively permeable membrane from a region of lower solute concentration to a region of higher solute concentration. In animal cells, an extracellular solution that is more concentrated than the cytoplasm causes the cell to lose water and shrink; a more dilute solution causes water to enter and the cell to swell."

Which is consistent with the passage?

- A) In a dilute external solution, an animal cell shrinks.
- B) Osmosis requires ATP.
- C) Water moves from high to low solute concentration.
- D) Water moves toward the higher-solute side.
- E) A concentrated external solution causes animal cells to swell.

2. "In genetics, a test cross is used to determine whether an individual displaying a dominant phenotype is homozygous or heterozygous. The individual is crossed with a homozygous recessive partner. If any offspring show the recessive phenotype, the tested individual must be heterozygous."

Which is the strongest inference?

- A) Test crosses require a homozygous dominant partner.
- B) A test cross always produces a 3:1 ratio.
- C) If all offspring show the dominant phenotype, the tested individual is definitely homozygous recessive.
- D) Test crosses can reveal the genotype only of recessive individuals.
- E) Observation of any recessive offspring proves the tested individual is heterozygous.

3. (General knowledge) The Italian chemist who proposed a number equal to the number of particles in one mole ($\sim 6.022 \times 10^{23}$), honoured in the constant's name though he did not measure it, is:

- A) Giulio Natta
- B) Amedeo Avogadro
- C) Enrico Fermi
- D) Stanislao Cannizzaro
- E) Mario Molina

4. Six boxes are stacked. Box R is above S but below T. Box U is immediately below S. Box V is at the very top. The order from top to bottom could be:

- A) T, V, R, S, U, W
- B) V, U, S, R, T, W
- C) V, T, R, S, U, (and W)
- D) V, R, T, S, U, W
- E) V, T, S, R, U, W

5. All squares are rectangles. Some rectangles are rhombi. Therefore:

- A) No valid conclusion about squares being rhombi follows.
- B) No square is a rhombus.
- C) Some rectangles are squares.
- D) All squares are rhombi.
- E) Some squares are rhombi.

6. If 5 machines make 5 widgets in 5 minutes, how long does it take 100 machines to make 100 widgets?

- A) 1 minute
- B) 5 minutes
- C) 20 minutes
- D) 100 minutes
- E) 500 minutes

7. What is the angle between the hands of a clock at 12:30?
A) 0° **B)** 15° **C)** 30° **D)** 165° **E)** 180°
8. Pointing to a photograph, Maria says, "Her mother is my mother's only daughter." How is the person in the photograph related to Maria?
A) Maria herself
B) Maria's mother
C) Maria's daughter
D) Maria's aunt
E) Maria's sister
9. A number is doubled, then 8 is added; the result is 20. The number is:
A) 4 **B)** 6 **C)** 8 **D)** 10 **E)** 12

Section B - Biology (Q10-32)

10. The mitochondrial matrix is the site of:
A) Lactate fermentation
B) Glycolysis
C) The light reactions of photosynthesis
D) The electron transport chain
E) The Krebs cycle
11. The net reaction of cellular respiration is:
A) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$
B) $C_6H_{12}O_6 \rightarrow 2 \text{ lactate}$
C) $C_6H_{12}O_6 \rightarrow 2 \text{ ethanol} + 2CO_2$
D) $6O_2 + 6CO_2 \rightarrow C_6H_{12}O_6$
E) $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
12. Which gas is produced by muscle cells during anaerobic respiration in yeast but NOT in humans?
A) CO_2 **B)** CH_4 **C)** N_2 **D)** O_2 **E)** H_2
13. The inner mitochondrial membrane is folded into:
A) Villi **B)** Microvilli **C)** Thylakoids **D)** Cisternae **E)** Cristae
14. The Calvin cycle of photosynthesis occurs in the:
A) Cytosol
B) Thylakoid membrane
C) Matrix
D) Stroma
E) Nucleus
15. Which is the correct site of the light-dependent reactions?
A) Stroma
B) Matrix
C) Thylakoid membrane
D) Cell wall
E) Cytosol
16. A solution of pH 2 is how many times more acidic than a solution of pH 5?
A) 3 times **B)** 10 times **C)** 30 times **D)** 100 times **E)** 1000 times

- 17.** The four levels of protein structure in order are:
A) Quaternary, tertiary, secondary, primary
B) Primary, tertiary, secondary, quaternary
C) Primary, secondary, quaternary, tertiary
D) Primary, secondary, tertiary, quaternary
E) Secondary, primary, tertiary, quaternary
- 18.** Which monosaccharide is the primary energy source for cells and the product of dietary carbohydrate digestion?
A) Fructose **B)** Galactose **C)** Glucose **D)** Lactose **E)** Sucrose
- 19.** Which of the following is a disaccharide?
A) Sucrose **B)** Glycogen **C)** Cellulose **D)** Fructose **E)** Glucose
- 20.** Lipids are largely insoluble in water because they are:
A) Highly polar
B) Non-polar (hydrophobic)
C) Composed only of amino acids
D) Charged
E) Always small molecules
- 21.** DNA differs from RNA in that DNA:
A) Has fewer nucleotide types
B) Is found only in the cytoplasm
C) Contains the sugar ribose
D) Contains thymine instead of uracil
E) Is usually single-stranded
- 22.** A population is in Hardy-Weinberg equilibrium for a locus with two alleles A (frequency p) and a (frequency q). The frequency of the heterozygote Aa is:
A) q^2 **B)** $2pq$ **C)** 1 **D)** $p + q$ **E)** p^2
- 23.** ABO blood groups are an example of:
A) Polygenic inheritance
B) Sex-linked inheritance
C) X-inactivation
D) Mitochondrial inheritance
E) Codominance (and multiple alleles)
- 24.** In humans, the normal diploid number of chromosomes is:
A) 23 **B)** 44 **C)** 46 **D)** 48 **E)** 92
- 25.** The cell-cycle phase in which DNA is replicated is:
A) S phase **B)** Cytokinesis **C)** G2 **D)** M phase **E)** G1
- 26.** Which vessels have thick muscular walls and carry blood away from the heart under high pressure?
A) Venules **B)** Arteries **C)** Veins **D)** Lymph vessels **E)** Capillaries
- 27.** The exchange of O_2 and CO_2 between blood and air occurs by:
A) Phagocytosis
B) Exocytosis
C) Active transport
D) Osmosis
E) Diffusion (passive)

28. The hormone secreted by the adrenal medulla in the "fight or flight" response is:
A) Adrenaline (epinephrine)
B) Thyroxine
C) Insulin
D) Oestrogen
E) Cortisol
29. The longest bone in the human body is the:
A) Tibia **B)** Humerus **C)** Femur **D)** Spine **E)** Skull
30. Smooth muscle is characterised by:
A) Multinucleation
B) Presence in the heart only
C) Branched cells with intercalated discs
D) Lack of striation and involuntary control
E) Striation and voluntary control
31. The organelle responsible for ATP production in animal cells is the:
A) Mitochondrion **B)** Nucleus **C)** Chloroplast **D)** Lysosome **E)** Ribosome
32. Which is the correct sequence of blood flow through the heart, starting from the body?
A) Vena cava → left atrium → left ventricle → aorta
B) Vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta
C) Aorta → right atrium → right ventricle → pulmonary vein
D) Right atrium → left atrium → right ventricle → left ventricle
E) Pulmonary vein → right atrium → right ventricle → pulmonary artery

Section C - Chemistry (Q33-47)

33. Calculate the number of moles in 22 g of CO₂. (C = 12, O = 16.)
A) 0.25 **B)** 0.5 **C)** 1 **D)** 2 **E)** 22
34. What volume does 0.25 mol of an ideal gas occupy at STP (V_m = 22.4 L/mol)?
A) 2.8 L **B)** 5.6 L **C)** 11.2 L **D)** 22.4 L **E)** 44.8 L
35. How many atoms are in 1.5 mol of He?
A) 1.20×10^{24} **B)** 1.50×10^{23} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 9.03×10^{23}
36. What is the molarity of a solution made by dissolving 5.85 g NaCl (molar mass 58.5) in water to make 500 mL?
A) 0.1 M **B)** 0.2 M **C)** 0.5 M **D)** 1.0 M **E)** 2.0 M
37. The oxidation number of Cr in Cr₂O₇²⁻ is:
A) -2 **B)** +3 **C)** +6 **D)** +7 **E)** +12
38. Identify the oxidising agent in: CuO + H₂ → Cu + H₂O.
A) Cu
B) H₂O
C) H₂ (it is oxidised)
D) CuO (it is reduced)
E) None
39. Calculate the pH of a 0.05 M solution of a strong base NaOH (assume full dissociation, 25 °C).
A) 1.3 **B)** 7 **C)** 12.7 **D)** 13 **E)** 14
40. The conjugate base of H₂O is:
A) H⁺ **B)** O₂ **C)** H₃O⁺ **D)** OH⁻ **E)** H₂

41. Which has the greatest first ionisation energy?
A) Ar **B)** Cl **C)** Na **D)** Al **E)** Mg
42. Which molecule has a linear geometry?
A) CH₄ **B)** H₂O **C)** SO₂ **D)** NH₃ **E)** CO₂
43. Which intermolecular force is strongest?
A) Hydrogen bonding
B) London dispersion
C) Van der Waals (general)
D) Dipole-dipole
E) Covalent bonding (intramolecular)
44. Increasing pressure on a gas at constant temperature will (Boyle's law):
A) Increase its volume
B) Increase its temperature
C) Decrease its volume
D) Have no effect
E) Decrease its moles
45. The structural formula CH₃-CH₂-OH represents:
A) Ethanoic acid **B)** Diethyl ether **C)** Propanol **D)** Ethanol **E)** Methanol
46. In the reaction: AgNO₃ + NaCl → AgCl + NaNO₃, the precipitate is:
A) NaCl
B) None — no precipitate forms
C) NaNO₃
D) AgNO₃
E) AgCl (insoluble)
47. For a reaction at equilibrium, adding a catalyst will:
A) Stop the reaction
B) Shift equilibrium toward reactants
C) Shift equilibrium toward products
D) Decrease K_{eq}
E) Increase the rate of both forward and reverse equally; equilibrium position unchanged

Section D - Mathematics (Q48-54)

48. Solve: $2x^2 - 8 = 0$.
A) No real solution
B) $x = \pm 4$
C) $x = \pm 2$
D) $x = 4$ only
E) $x = 2$ only
49. Solve: $|x - 3| = 5$.
A) $x = 2$
B) $x = -2$ only
C) $x = 8$ only
D) $x = 8$ or $x = -2$
E) $x = 5$
50. Solve: $\log_2(x) + \log_2(x - 3) = 2$ ($x > 3$).
A) $x = 4$ **B)** $x = -1$ **C)** $x = 3$ **D)** $x = 8$ **E)** $x = 1$

51. Solve the inequality: $x^2 - 4 < 0$.

- A) $x < -2$ B) $x < 2$ C) $x \neq 2$ D) $x > 2$ E) $-2 < x < 2$

52. A right circular cylinder has radius 3 and height 10. What is its volume? (Use $\pi = 3.14$.)

- A) 30 B) 94.2 C) 282.6 D) 314 E) 942

53. $\tan 45^\circ = ?$

- A) 1 B) $\sqrt{3}$ C) 0 D) $\sqrt{3}/3$ E) $1/2$

54. A die is rolled once. What is the probability of an even number greater than 2?

- A) $1/6$ B) $1/3$ C) $1/2$ D) $2/3$ E) $5/6$

Section E - Physics (Q55-60)

55. A car accelerates from 5 m/s to 25 m/s in 10 s. What is its acceleration?

- A) 1 m/s^2 B) 2 m/s^2 C) 5 m/s^2 D) 20 m/s^2 E) 25 m/s^2

56. A 12 V battery drives a current through a 4Ω resistor in series with a 2Ω resistor. What is the current?

- A) 1 A B) 2 A C) 3 A D) 6 A E) 12 A

57. A 3 A current flows for 2 minutes. What total charge passes a point?

- A) 6 C B) 60 C C) 180 C D) 360 C E) 600 C

58. A 2 kg object moving at 3 m/s collides with a stationary 1 kg object and they stick together. What is their common velocity?

- A) 1 m/s B) 2 m/s C) 3 m/s D) 4 m/s E) 6 m/s

59. A 4 kg object is dropped (from rest). Ignoring air resistance, what is its velocity after 3 s? ($g = 9.8 \text{ m/s}^2$.)

- A) 9.8 m/s B) 19.6 m/s C) 29.4 m/s D) 39.2 m/s E) 88.2 m/s

60. A wave has frequency 50 Hz and wavelength 4 m. What is its speed?

- A) 12.5 m/s B) 50 m/s C) 100 m/s D) 200 m/s E) 400 m/s

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	D	2	E	3	B	4	C	5	A
6	B	7	D	8	C	9	B	10	E
11	A	12	A	13	E	14	D	15	C
16	E	17	D	18	C	19	A	20	B
21	D	22	B	23	E	24	C	25	A
26	B	27	E	28	A	29	C	30	D
31	A	32	B	33	B	34	B	35	E
36	B	37	C	38	D	39	C	40	D
41	A	42	E	43	A	44	C	45	D
46	E	47	E	48	C	49	D	50	A
51	E	52	C	53	A	54	B	55	B
56	B	57	D	58	B	59	C	60	D